Abhishek Dhanani

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Education

Sacred Heart University Fairfield, CT
Master's in computer science – data science track ***GPA: 3.9*** *Aug. 2024 – Dec. 2025*
- Coursework: Machine learning, Deep Learning, Big Data Analysis

Government Engineering College Gandhinagar, Gujarat, India
Bachelor's in computer science *Aug. 2020 – May 2023*
- Coursework: Algorithms, Database Management, Advanced Mathematics, Software Engineering

Technical Skills

Programming Languages: Python, SQL, Java
Data Visualization: Tableau, Matplotlib, Seaborn
Developer Tools: Git, Visual Studio, PyCharm, Jupyter notebook
Machine Learning Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers

Experience

Data analysis intern Jan 2023 – July 2023
Infolabz Pvt. Ltd. *Ahmedabad, INDIA*
- Cleaned and analyzed data using Pandas and NumPy to identify trends, patterns and actionable insights using tools like excel and Python.
- Developed visualizations and reports with Tableau and Matplotlib to present findings to teams.
- Contributed to the development of regression and classification models with Scikit-learn, to gain meaningful insights about data and analyze the market trends.
- Worked closely with marketing team to gather data requirements and deliver insights that supported strategic decision-making resulting in sales improvement.

Projects

NLP Exploration of United Nations Discourse | NLP, NLTK, spaCy, *Python, LDA, NFA* *Dec 2025*
- Analyzed over 4,700 UN General Debate speeches (2000-2024) using **NLP techniques** for tokenization, word frequency analysis, sentiment analysis, and **topic modeling (LDA/NMF)** to uncover trends in global discourse on climate, security, and development.
- Generated visualizations including heatmaps, word clouds, and trend charts to highlight shifts (e.g., increased focus on sustainability post-2015 and regional variations across G7, G20, EU, and Africa).
- **Paper accepted** to the 2026 International Conference on Breakthroughs in Artificial Intelligence **(BAI'26)**; to be published in Springer Lecture Notes in Networks and Systems (LNNS).
- Demonstrated expertise in data preprocessing, machine learning frameworks, vector databases, and interactive tools (pyLDAvis).

MailCast: Email to Audio Generation | *NLP, Python, PyDub, OpenAI API, Gemini API* *Mar 2025*
- Developed a **Python application** to convert emails into audio podcasts by integrating **Gmail API** for email retrieval and BeautifulSoup for **text preprocessing**.
- Utilized a large language model **(LLM)** to generate podcast scripts from email content and **OpenAI's Text-to-Speech (TTS) API** with **PyDub** for audio output.
- Implemented **OAuth2 authentication** and chunked **text processing** to handle API token limits, ensuring scalability for large email datasets.
- Demonstrated skills in API integration, text processing, natural language generation, and audio generation.

RAG System for Document Summarization | *Python, LangChain, ChromaDB, OpenAI Embedding, GPT models* *Mar 2025*
- Designed a Retrieval-Augmented Generation **(RAG)** system in Python to summarize lengthy legal documents, preprocessing text and storing summaries in a **Chroma vector database**.
- Integrated **OpenAI embeddings** and GPT models with **LangChain** to retrieve and generate concise, accurate summaries from document sections.
- Built **a multi-vector retriever pipeline** to combine original text and summaries, enhancing retrieval efficiency and summary coherence.
- Showcased expertise in NLP, vector databases, retrieval techniques, and large language models.

Fake News Classification using DistilBERT | *NLP, Python, Hugging Face Transformers* *Spring 2025*

- Developed an **NLP model** to classify news articles as real or fake by **fine-tuning DistilBERT** using **Hugging Face's Transformers** library and **AutoTokenizer** for efficient text preprocessing.
- Processed a dataset of over **44,000 articles** (Fake.csv and True.csv), performing data cleaning, tokenization, and splitting (70% train, 15% validation, 15% test).
- Fine-tuned DistilBERT on **PyTorch** with GPU, leveraging mixed precision training (FP16) for faster computation and reduced memory usage. Achieving an accuracy of 99.98% on the test set, with a confusion matrix and classification report for evaluation.
- Demonstrated skills in NLP, deep learning, model optimization, and Hugging Face ecosystem integration.